Parts and Functions of a Compound Microscope

Light Microscope

- Simple – uses a single lens
- Compound – uses a set of lenses or lens systems

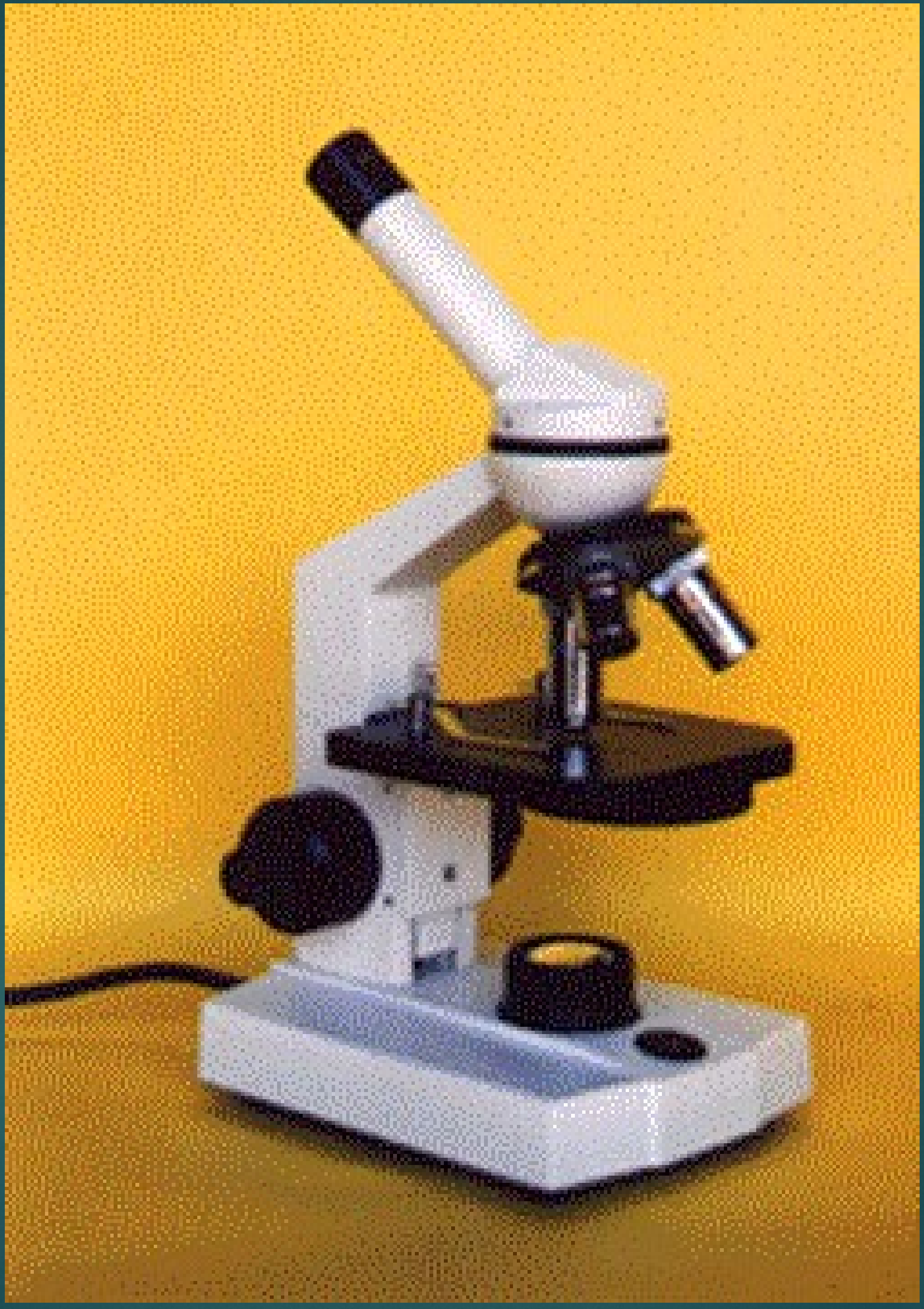
Simple Light Microscope

Define the ff: (see module)

- Specimen
- Slide
- Cover slip
- Field of View
- Parfocal
- Magnification
- Resolution

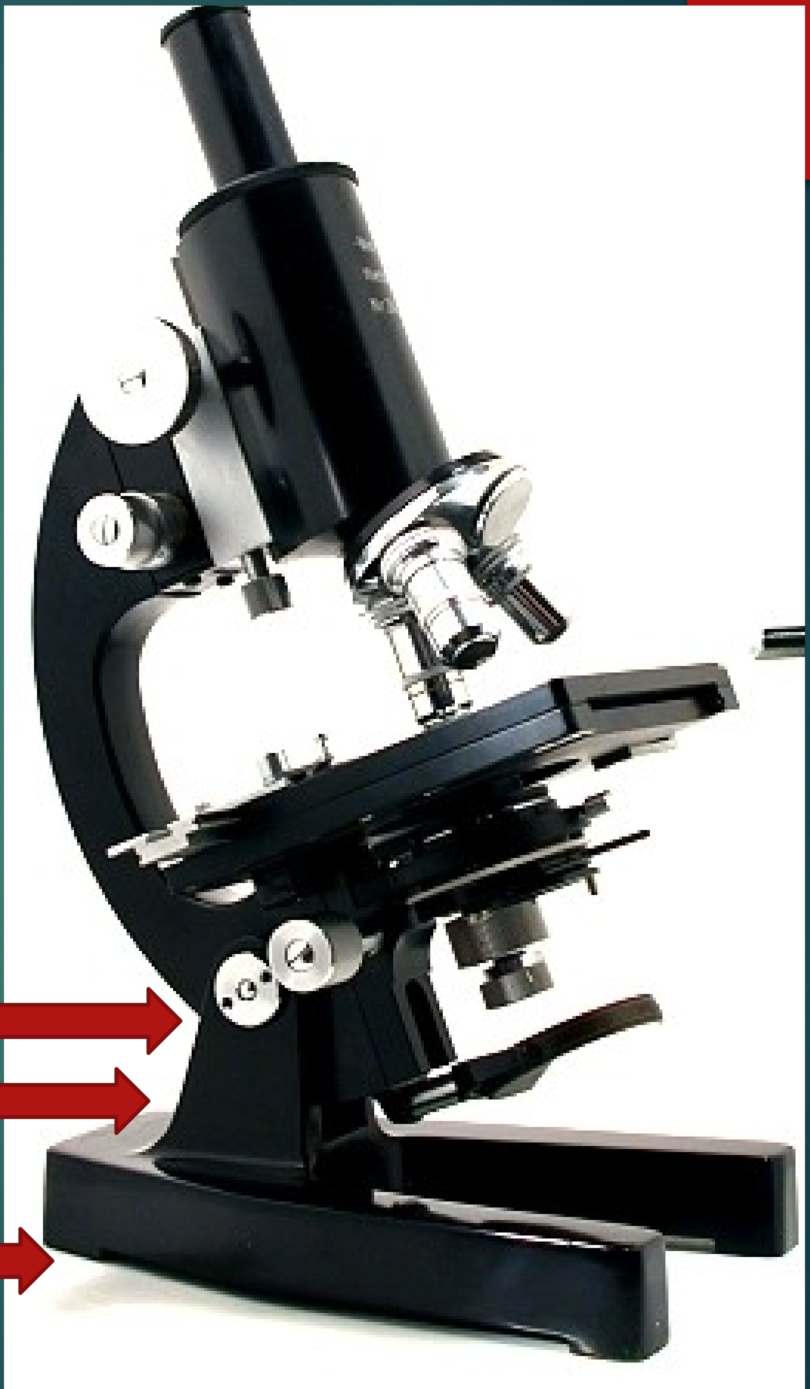
Compound Microscope

- **Mechanical Parts**
  - Used to support and adjust the parts
- **Magnifying Parts**
  - Used to enlarge the specimen
- **Illuminating Parts**
  - Used to provide light



Mechanical Parts

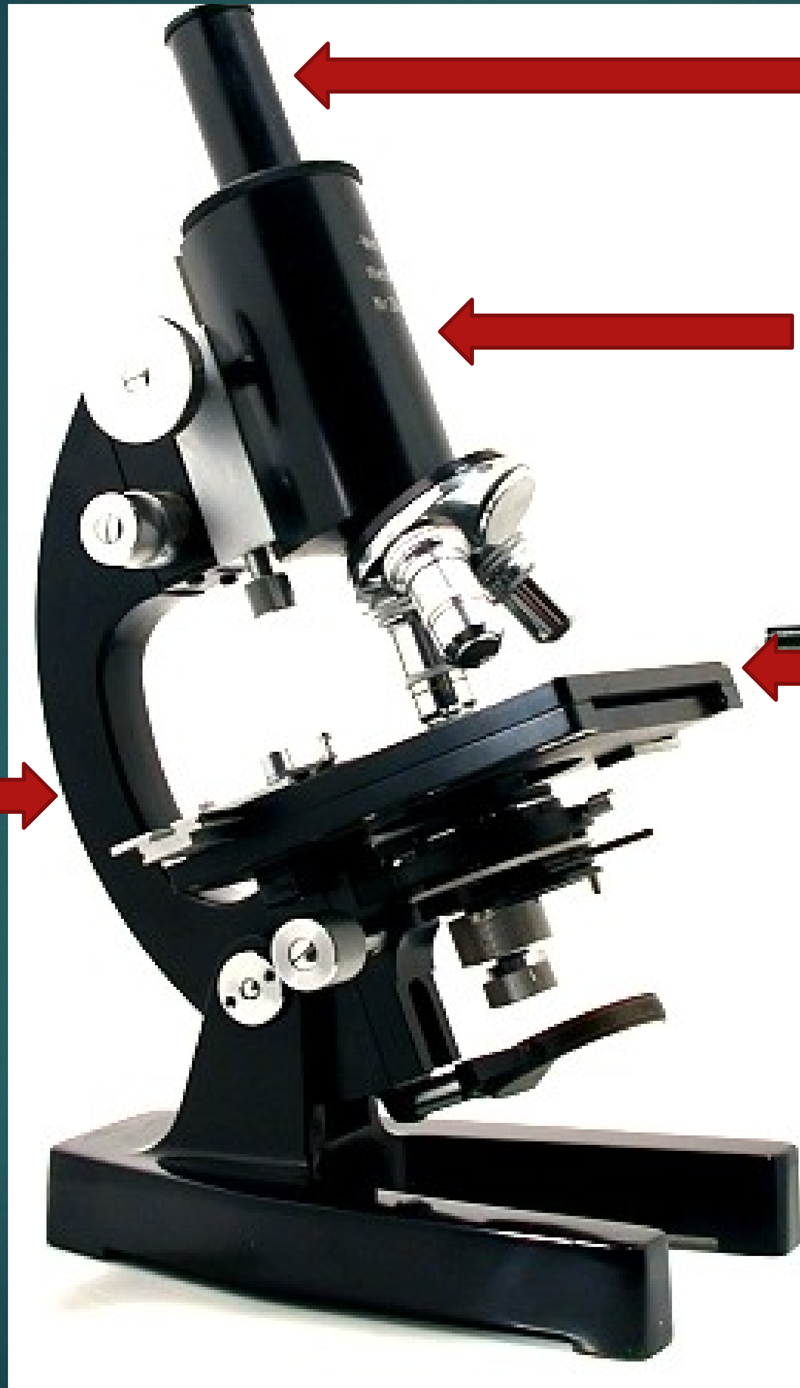
- Base
  - Bottommost portion that supports the entire/lower microscope
- Pillar
  - Part above the base that supports the other parts
- Inclination Joint
  - Allows for tilting of the microscope for convenience of the user



Inclination Joint
Pillar
Base

- Arm/Neck
  - Curved/slanted part which is held while carrying the microscope
- Stage
  - Platform where object to be examined is placed
- Stage Clips
  - Secures the specimen to the stage

- Stage Opening
- Body Tube
  - Attached to the arm and bears the lenses
- Draw Tube
  - Cylindrical structure on top of the body tube that holds the ocular lenses


Draw Tube
Body Tube
Stage
Arm /
Neck

- Revolving/Rotating Nosepiece
  - Rotating disc where the objectives are attached

- Dust Shield
  - Lies atop the nosepiece and keeps dust from settling on the objectives

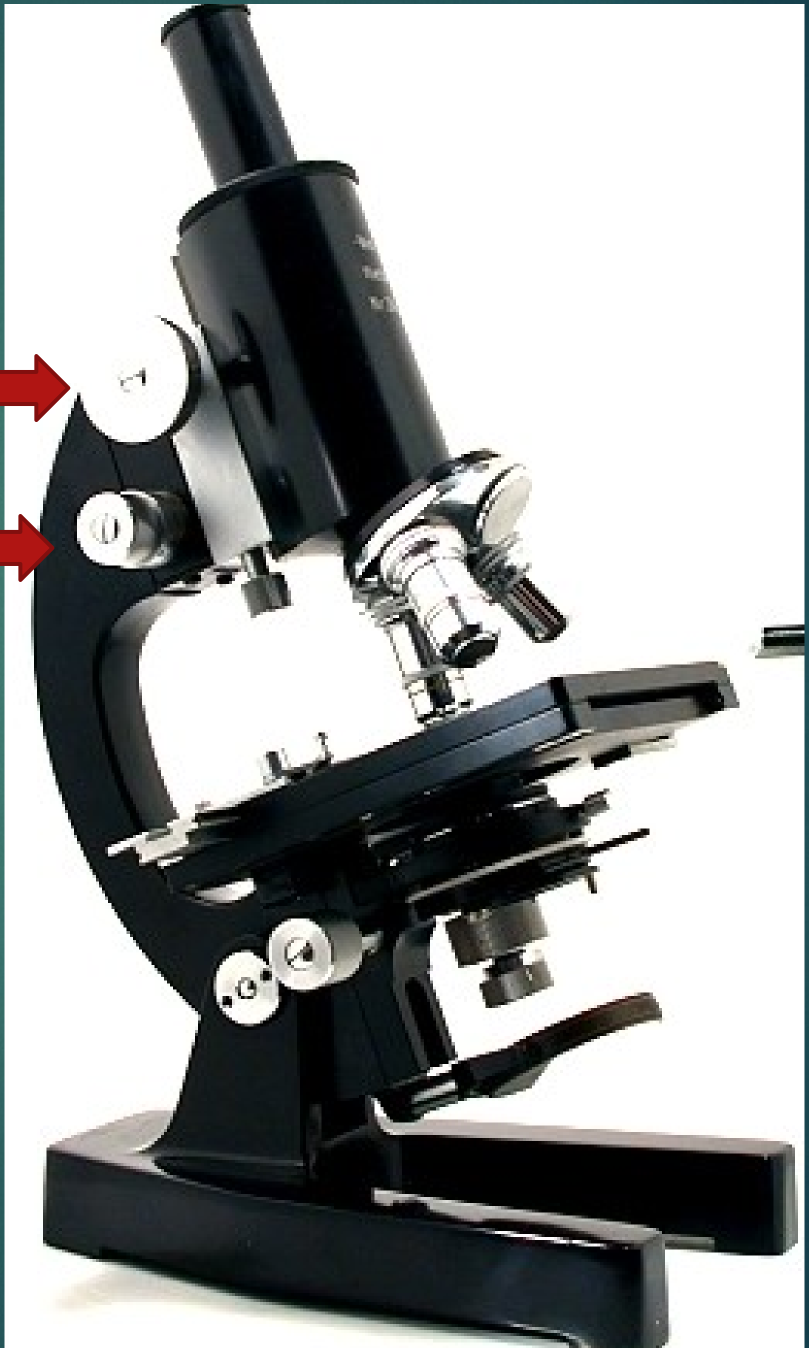
Revolving
Nosepiece

- Coarse Adjustment Knob
  - Geared to the body tube which elevates or lowers when rotated bringing the object into approximate focus

- Fine Adjustment Knob
  - A smaller knob for delicate focusing bringing the object into perfect focus

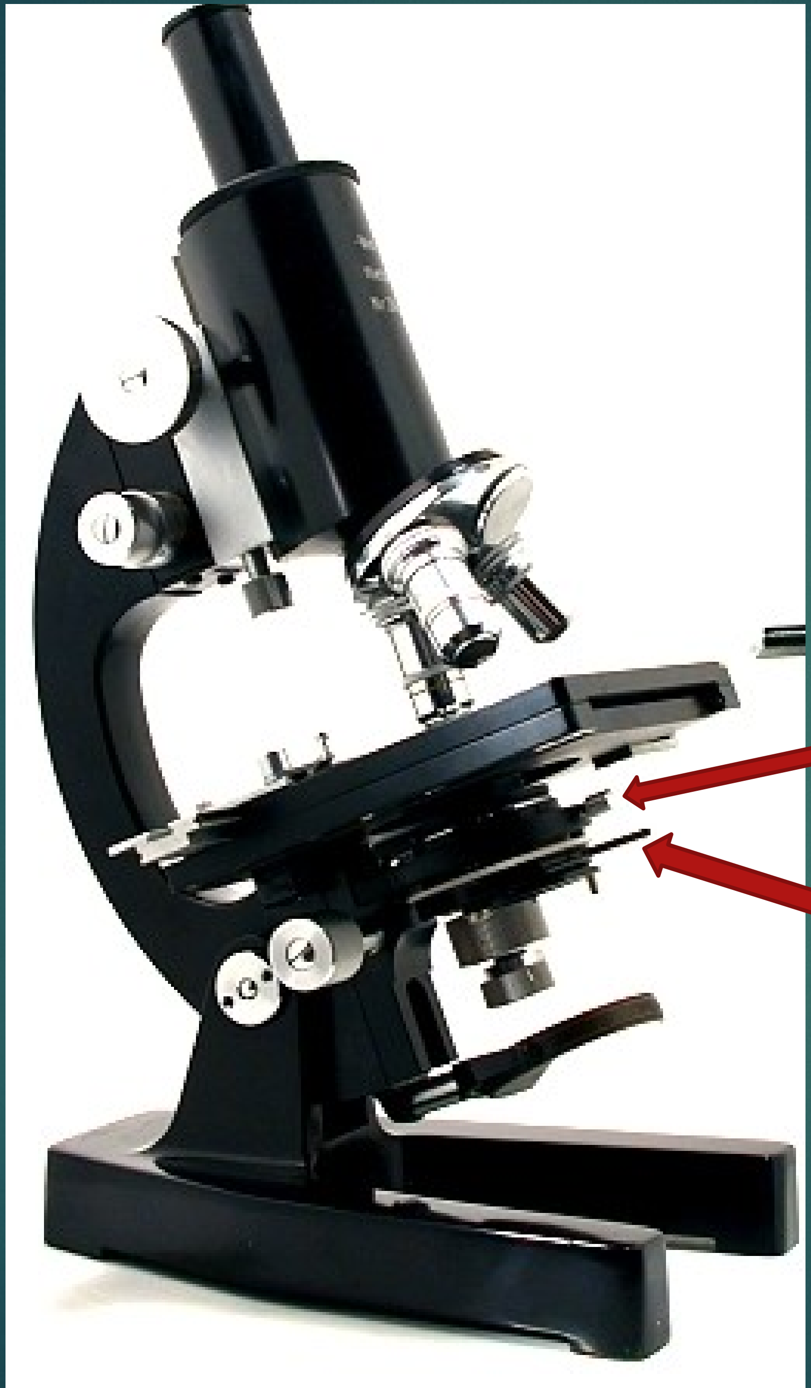
Coarse
Adjustment
Knob

Fine
Adjustme
nt Knob



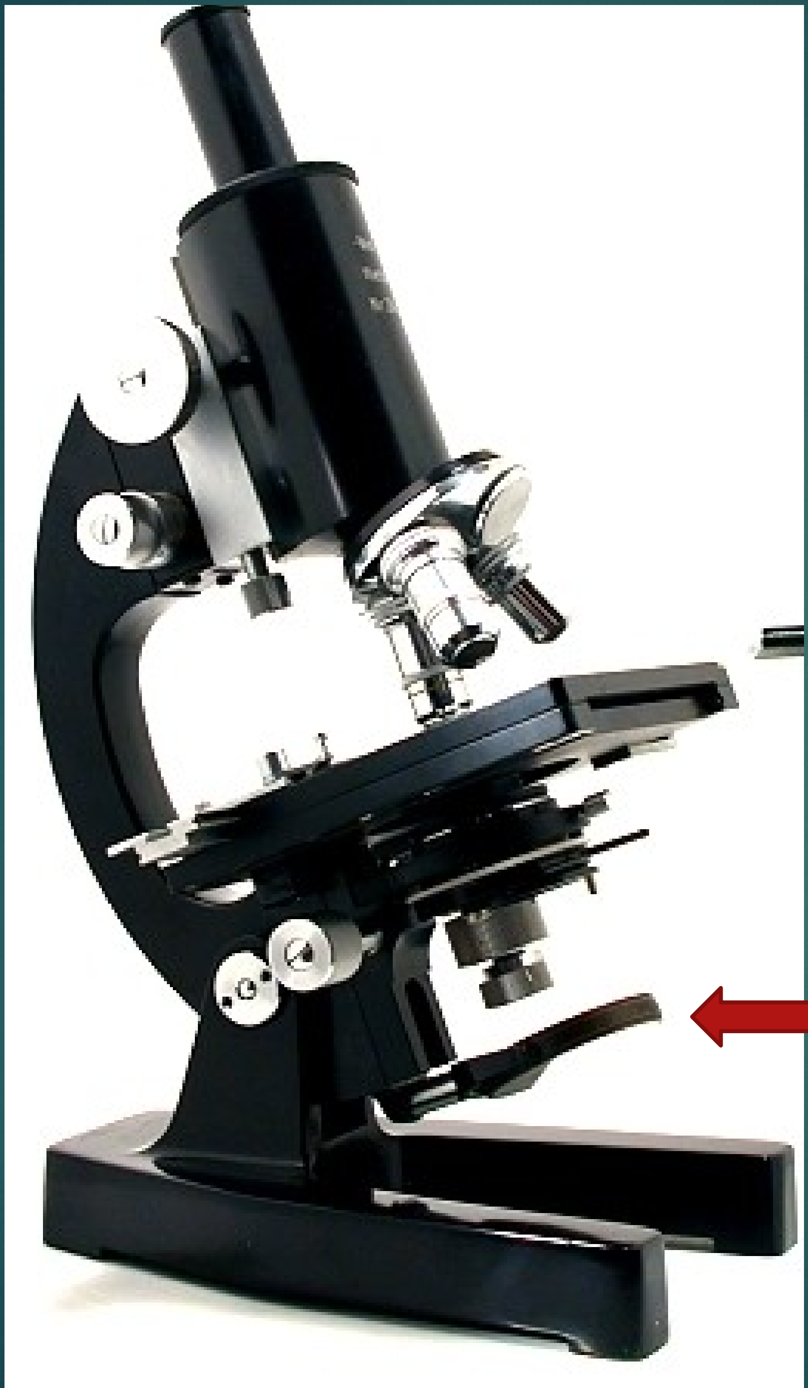
- Condenser Adjustment Knob
  - Elevates and lowers the condenser to regulate the intensity of light

- Iris Diaphragm Lever
  - Lever in front of the condenser and which is moved horizontally to open/close the diaphragm


Iris Diaphragm
Lever
Condenser
Adjustment Knob

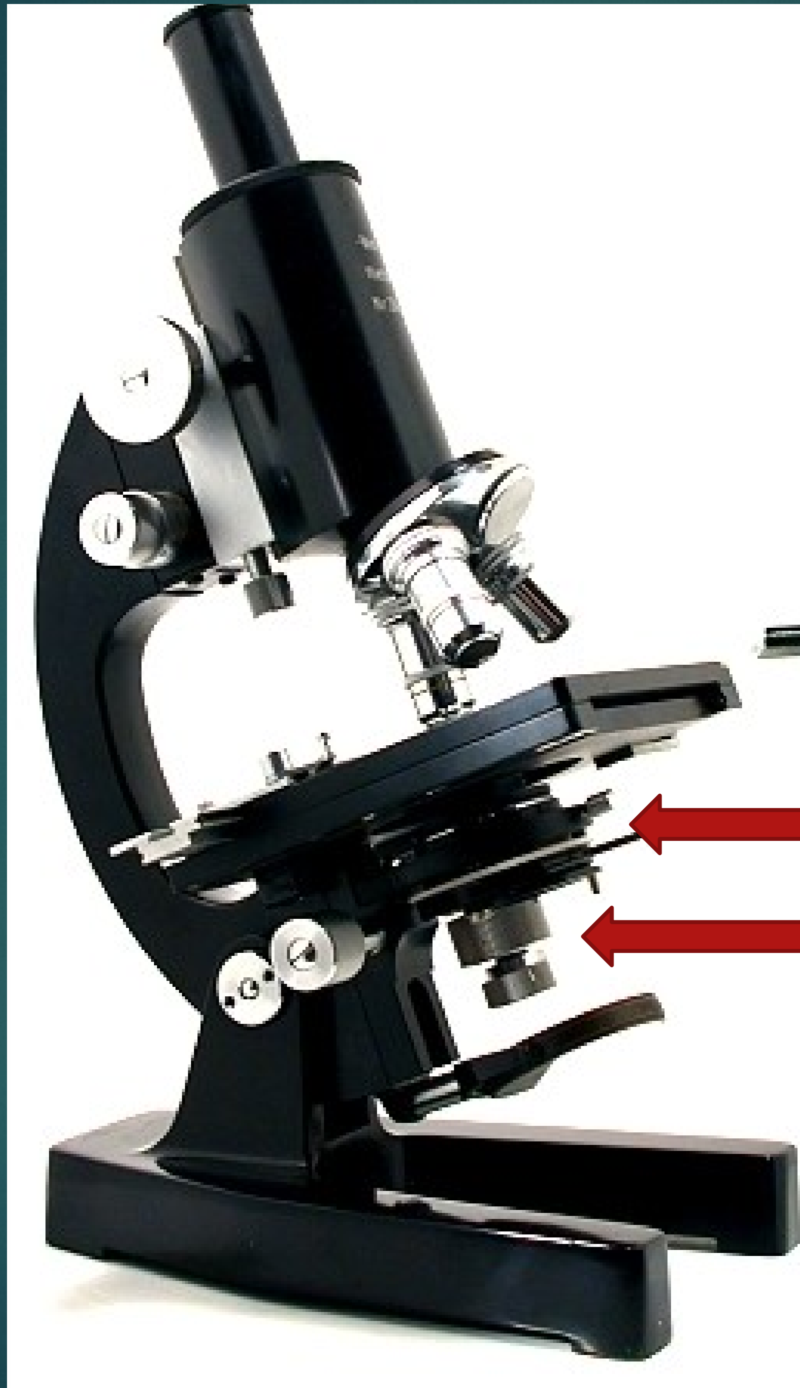
Illuminating Parts

- Mirror
  - Located beneath the stage and has concave and plane surfaces to gather and direct light in order to illuminate the object
- Electric Lamp
  - A built-in illuminator beneath the stage that may be used if sunlight is not preferred or is not available



Mirror / Electric Lamp

- Substage
  - Iris Diaphragm
    - Regulates the amount of light necessary to obtain a clearer view of the object
  - Condenser
    - A set of lenses between the mirror and the stage that concentrates light rays on the specimen.


Iris Diaphragm
Condenser

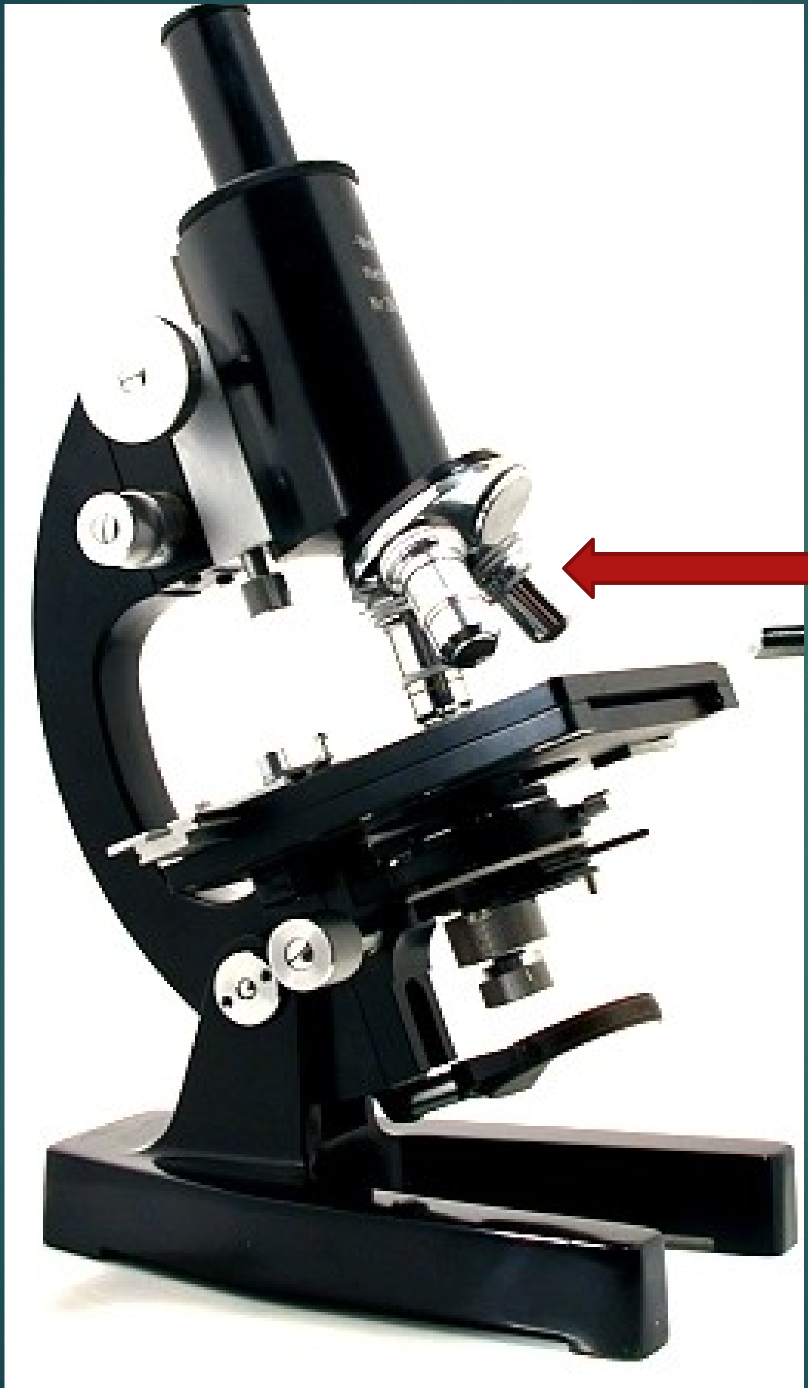
MAGNIFYING PARTS

- Ocular / Eyepiece
  - Another set of lens found on top of the body tube which functions to further magnify the image produced by the objective lenses. It usually ranges from 5x to 15x.


Ocular

MAGNIFYING PARTS

- Objectives
  - Metal cylinders attached below the nosepiece and contains especially ground and polished lenses
    - LPO / Low Power Objective
      - Gives the lowest magnification, usually 10x
    - HPO / High Power Objective
      - Gives higher magnification usually 40x or 43x
    - OIO / Oil Immersion Objective
      - Gives the highest magnification, usually 97x or 100x, and is used wet either with cedar wood oil or synthetic oil



Objectives

Use of the Compound Microscope

- Make sure all backpacks are **out of the aisles** before you get a microscope!

- Always carry the microscope with one hand on the **Arm** and one hand on the **Base.** Carry it **close to your body.**

- Be gentle.
- Setting the microscope down on the table roughly could jar lenses and other parts loose.

- Always start and end with l**owest powered objective.**

- Place the slide on the microscope stage, with the specimen directly over the **center** of the glass circle on the stage (directly over the light).

- *If you wear **glasses**, take them **off**; if you see only your **eyelashes**, move **closer.***

- If you see a **dark line** that goes *part way* across the field of view, try turning the eyepiece.

- Use only the **Fine adjustment knob** when using the **HIGH** (long) **POWER OBJECTIVE.**

- As much as possible, keep both eyes open to reduce eyestrain. Keep eye slightly above the eyepiece to reduce eyelash interference.

- If, ***and ONLY if***, you are on ***LOW POWER***, lower the objective lens to the **lowest point**, then focus using first the coarse knob, then the fine focus knob.

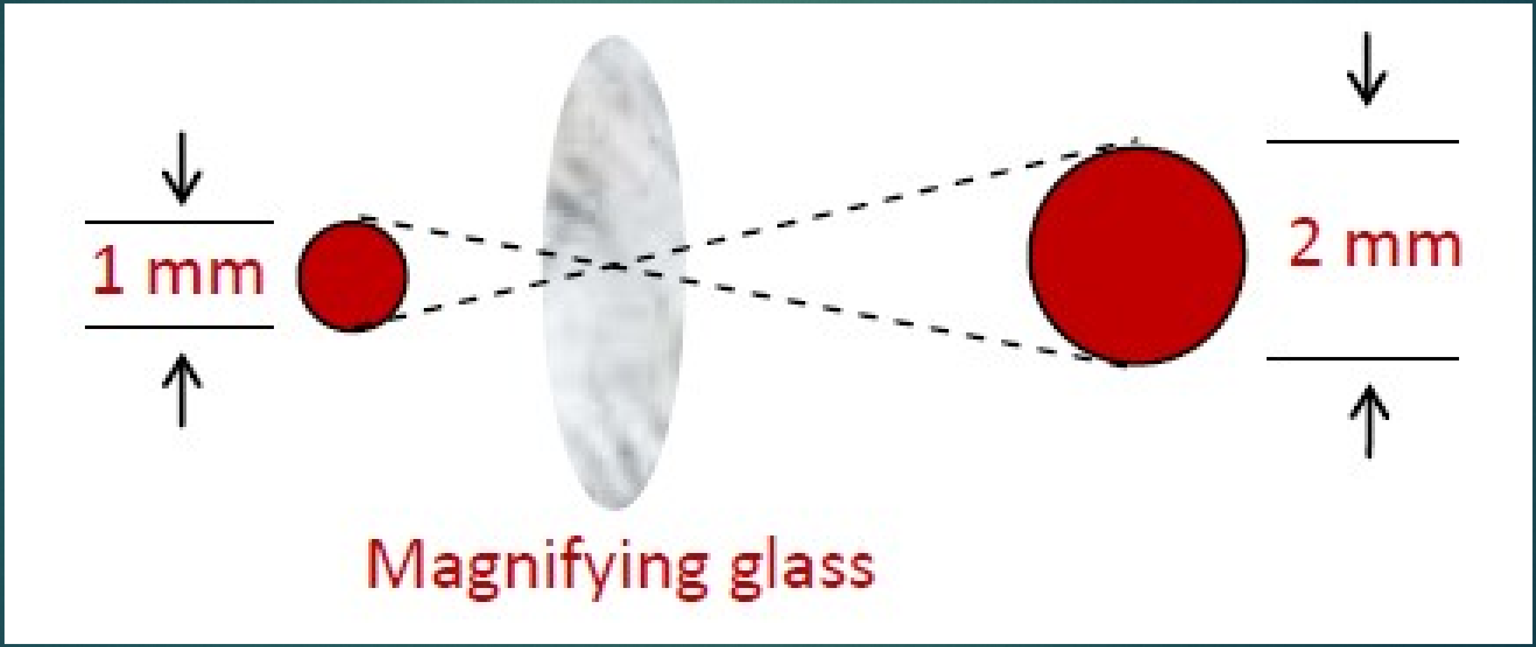
- Adjust the **Diaphragm** as you look through the **Eyepiece**, and you will see that **MORE** detail is visible when you allow in **LESS** light!
- ***Too much*** light will give the specimen a ***washed-out*** appearance.

- Once you have it on **High Power** remember that you **only use the fine focus knob!**

- The High Power Objective (40x) is **very close** to the slide. Use of the coarse focus knob will **scratch** the lens, and **crack** the slide.

MAGNIFICATION

- The ratio of the original image to the “magnified” image.



RESOLUTION

- limiting distance between two points at which they are perceived as distinct from one another.

Numerical Aperture

- the amount of light that which enters the objective.
- The larger the NA, the greater the resolving power of the objective.

Objective Specifications		
Magnification	Numerical Aperture	Resolution, *u*m
5x	0.13	2.1
10x	0.25	1.1
20x	0.5	0.55
50x	0.8	0.34
100x	0.9	0.31